



KHETMITRA

THEME: AI-POWERED SMART AGRICULTURE

- **Problem Statement Title** - AI-powered Smart Agriculture Advisor: boosting Crop Yields and Farmer Decision-Making
- **Theme** - AI for Social Impact (Agriculture)
- **Team Name** - AI (Aloo Intelligence)
- **Team Member** - Nitish Sheoran (Team Leader)
Narayan Prasad
Sonal Tyagi

Website:- <https://www.khetmitra.live/>





KHETMITRA : AI-Powered Personal Farming Assistant for Farmers

Your AI-powered farming companion for smarter, sustainable harvests

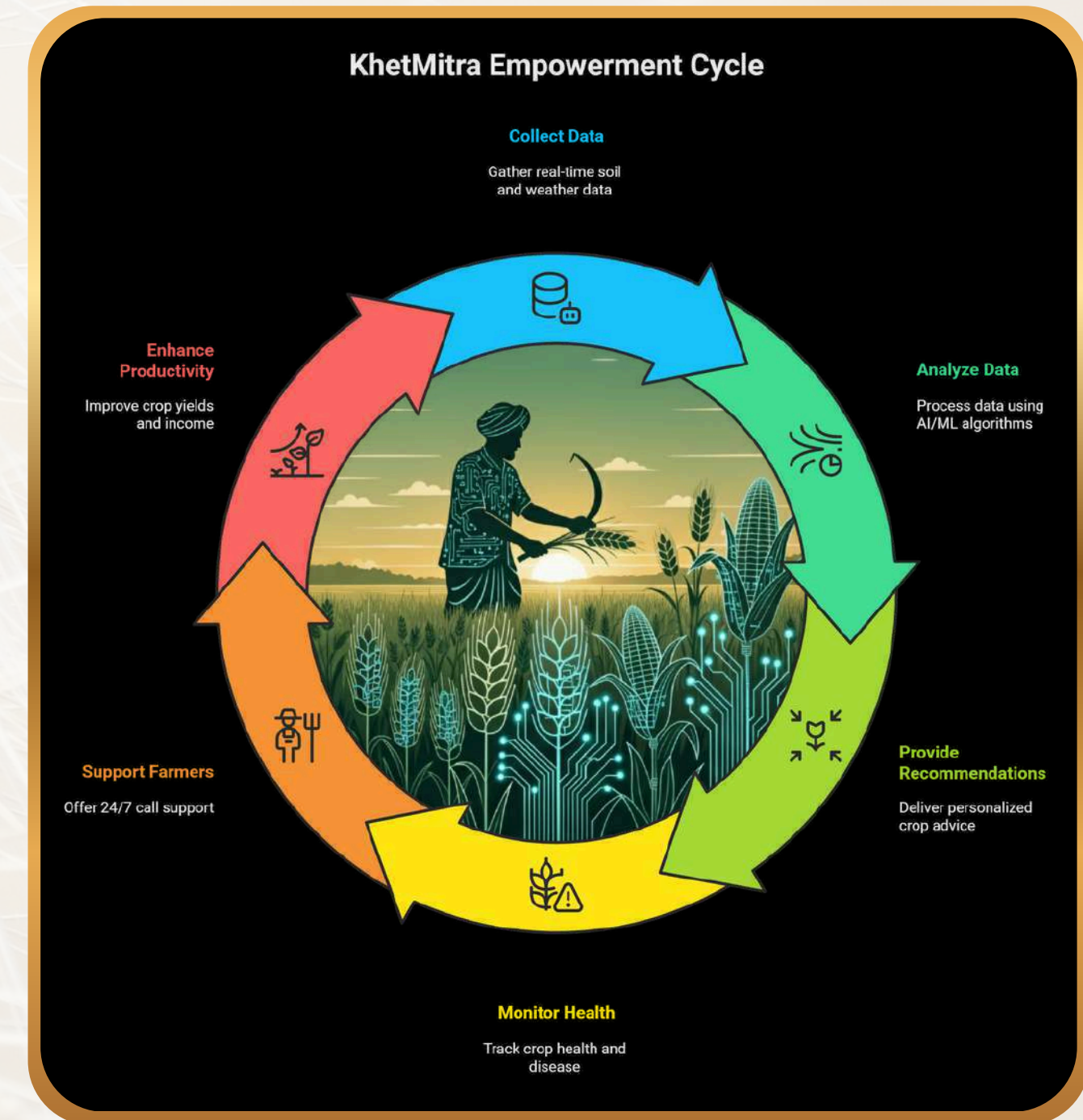
DETAILED EXPLANATION OF THE PROPOSED SOLUTION

KhetMitra uses soil sensors with an **ESP32 and SIM module** to collect real-time soil data, combined with weather, satellite data, soil maps, crop history, and current mandi bhav. AI/ML analyzes this data to recommend crops, predict yield and profit, monitor crop health, detect diseases, and support sustainable farming. The platform offers an **“end-to-end solution—from crop selection to selling”**—helping farmers make smarter decisions and boost productivity. It supports multiple languages, works offline for areas with poor connectivity, and provides 24/7 call support to assist farmers anytime.

INNOVATION & UNIQUENESS OF THE SOLUTION



HOW IT ADDRESSES THE PROBLEM





TECHNICAL APPROACH

TECHNOLOGIES TO BE USED

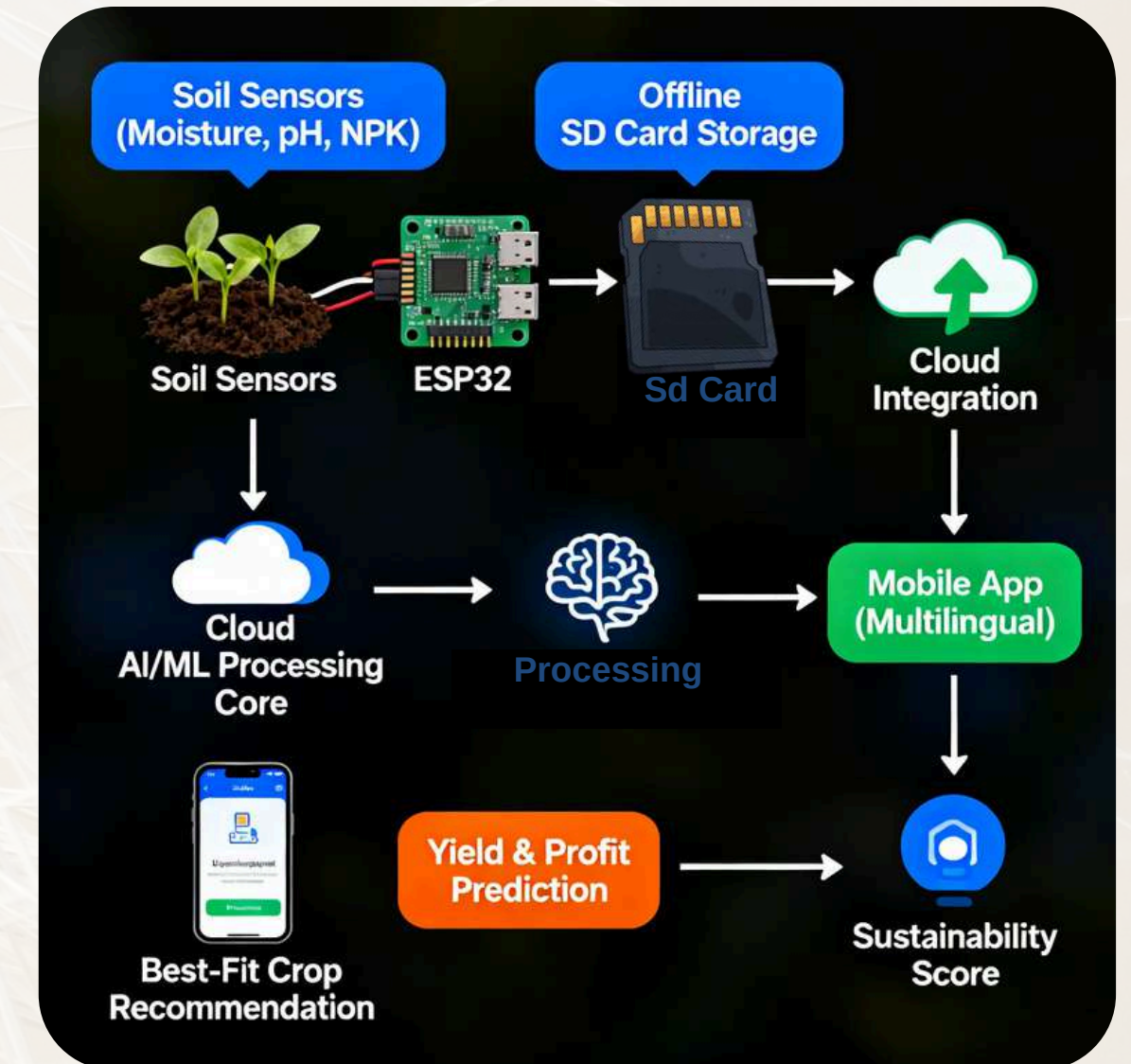
SOFTWARE:



HARDWARE:

- Soil Sensors (pH, Moisture, Nutrients)
- ESP32 Microcontroller
- SD Card Module
- IoT connectivity (Wi-Fi&4G).

METHODOLOGY & PROCESS OF IMPLEMENTATION



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FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS

- ✓ Cost-effective at ₹4,500
- ✓ Government policy alignment (PM-KISAN)
- ✓ Multiple revenue streams & scalable model



CHALLENGES & RISKS

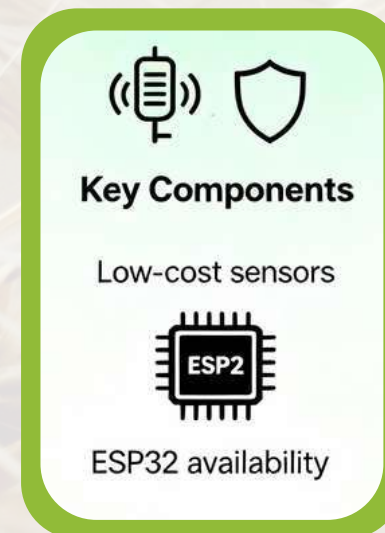
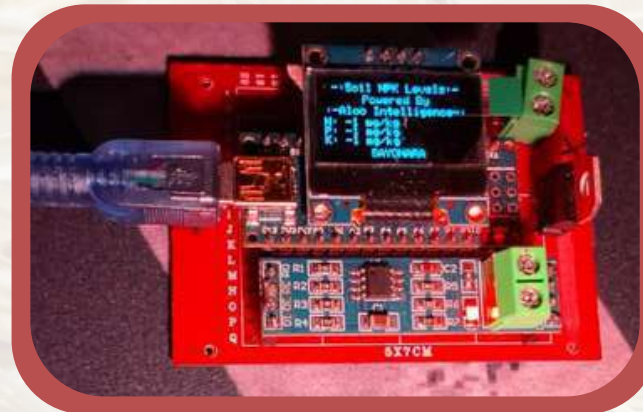
- Calibration accuracy
- Rural connectivity gaps
- Data privacy issues



STRATEGIES

- Field validation + calibration
- Offline data logging (ESP32 + SD card)
- Secure APIs + cloud encryption

PROTOTYPE IMAGES

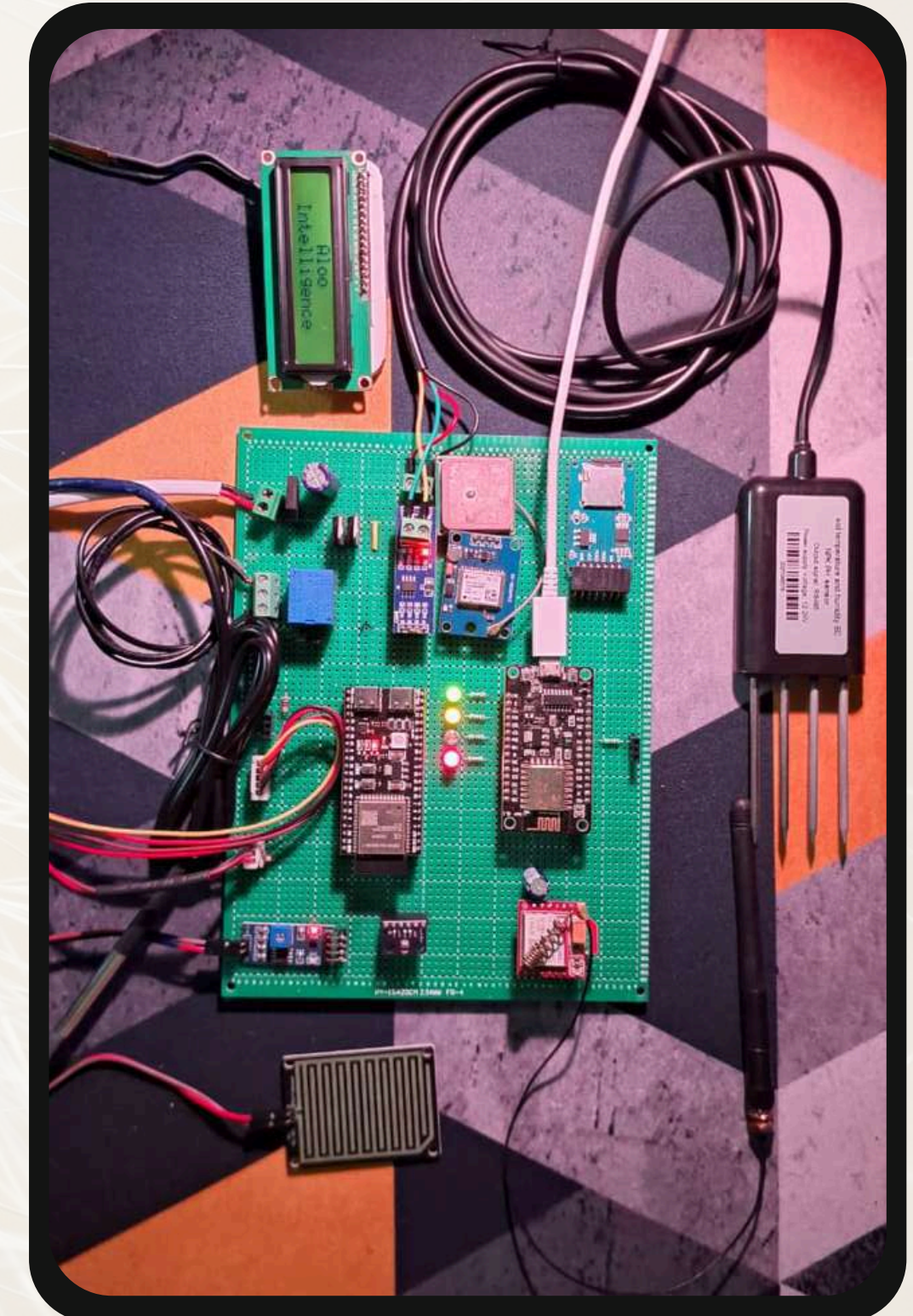
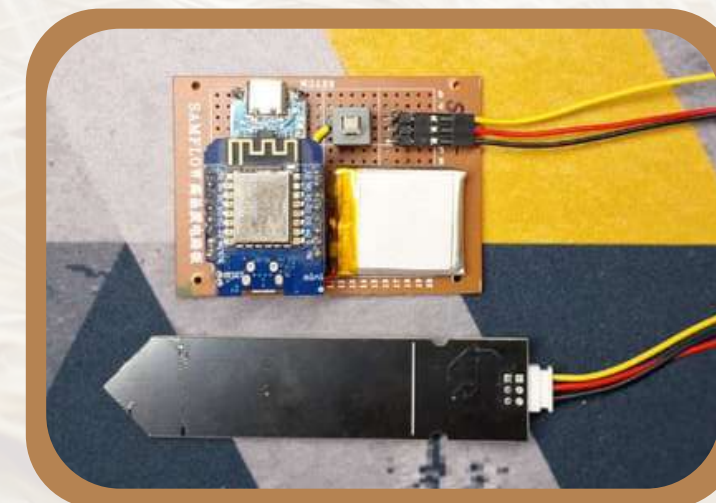


Key Components

Low-cost sensors



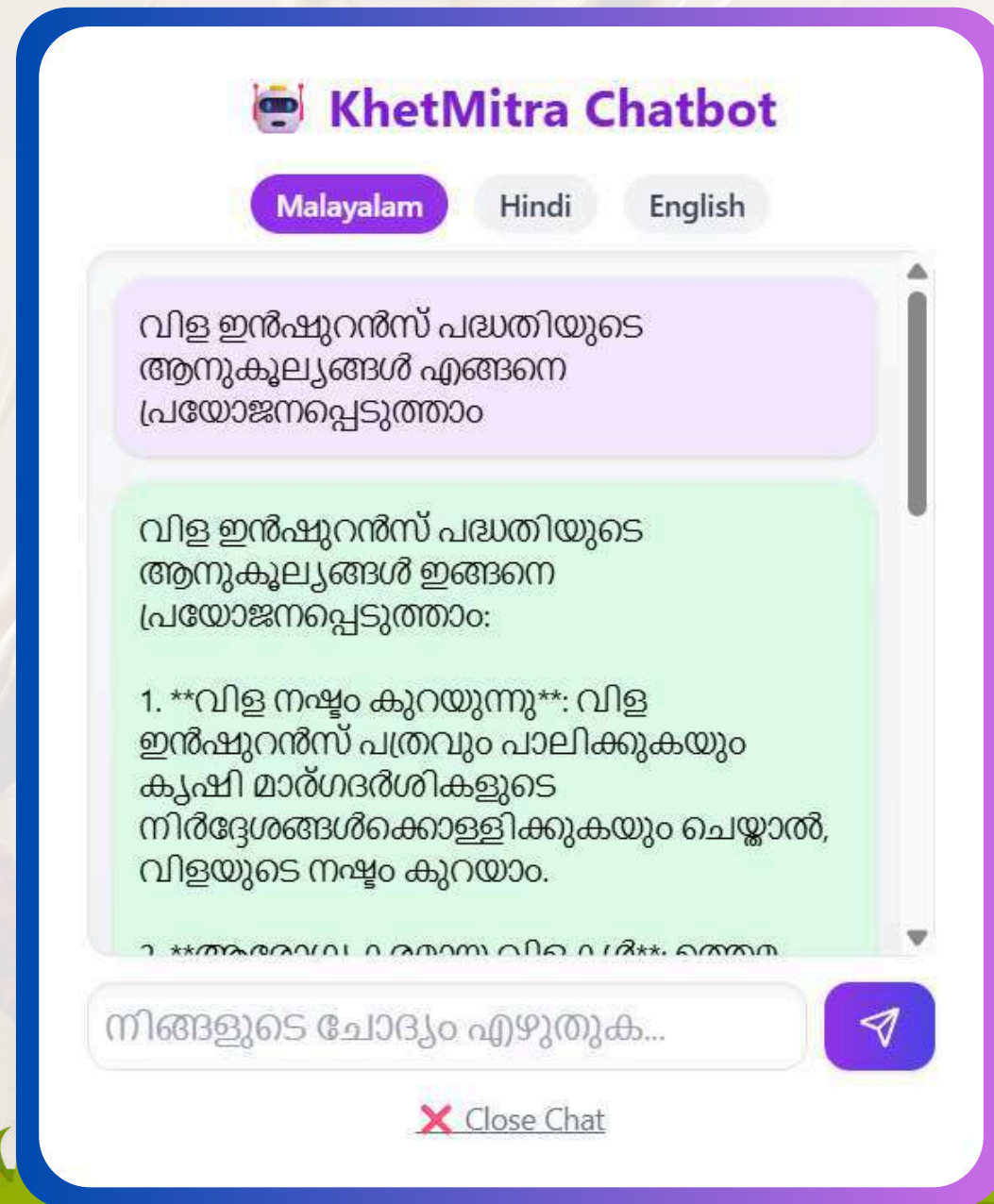
ESP32 availability





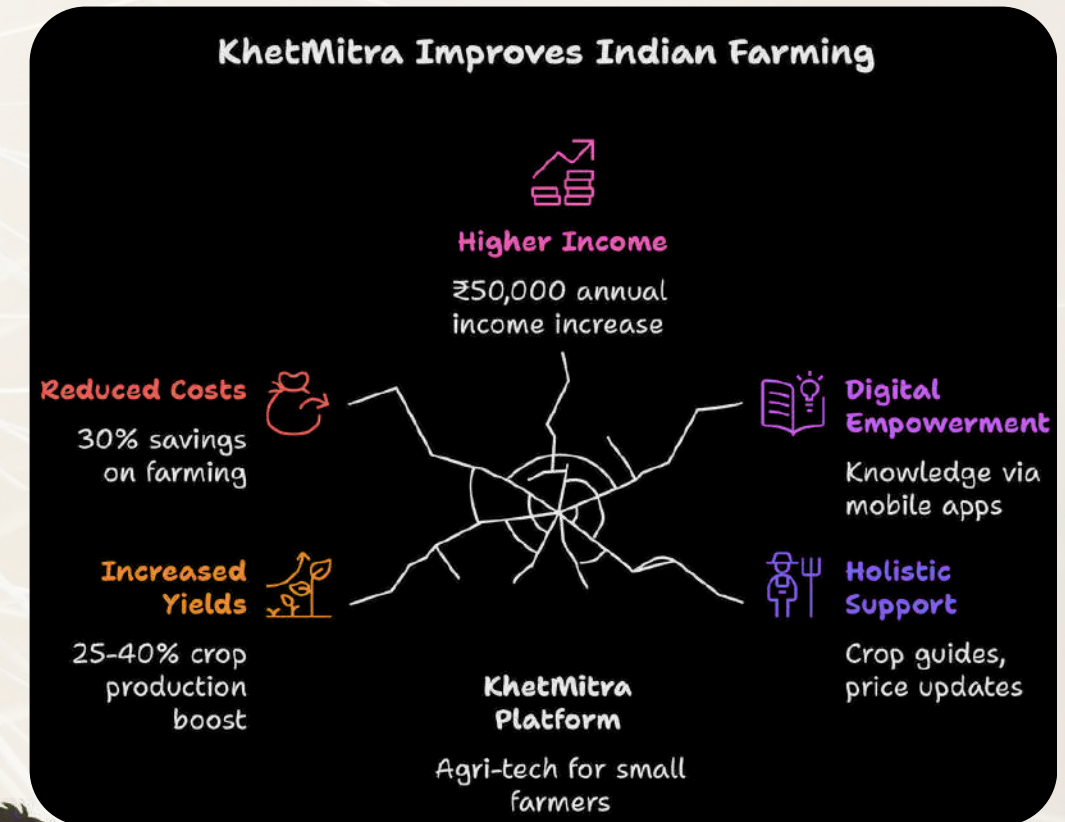
IMPACT AND BENEFITS

BENIFIT OF THE SOLUTION



POTENTIAL IMPACT

- Data-driven crop recommendations & yield prediction
- AI-backed insights replacing guesswork
- Smarter irrigation & fertilizer planning
- Real-time alerts (weather, pests, soil health, Grazing & High wind)
- Market linkage support for better income





RESEARCH AND REFERENCES

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3. Crop recommendation system for growing best suitable crop International Journal of Science and Research Archive, 2024, 12(01), 2928–2936 Publication history: Received on 08 May 2024; revised on 15 June 2024; accepted on 18 June 2024 Article DOI: <https://doi.org/10.30574/ijrsra.2024.12.1.1111>
4. Machine learning based recommendation of agricultural and horticultural crop farming in India under the regime of NPK
5. <https://pmkisan.gov.in/>
6. <https://www.digitalindia.gov.in/initiative/pm-kisan/>
7. <https://pmfby.gov.in>
8. Crop variety management for climate adaptation supported by citizen science

